

RANIYA IQBAL

ROBOTICS AND AUTOMATION ENGINEER

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PROFILE

Mechatronics Engineer specializing in robotics, automation, and embedded AI systems — from autonomous inspection robots to AI-driven wearable health tech. Hands-on expertise across sensor fusion and computer vision (LiDAR, OpenCV), embedded programming (Arduino, ESP32, Raspberry Pi, micro:bit), PLC and industrial automation, and CAD/mechanical design (SolidWorks, Fusion 360, ANSYS). Experienced in machine learning model development (Random Forest, SHAP explainability) and foundational AI agent concepts, paired with PLC/ELV field experience including fibre-optic diagnostics and OTDR testing. SAP S/4HANA-certified, with a track record of leading cross-functional teams from concept through working prototype and thriving under deadline pressure.

EMPLOYMENT HISTORY

Emerson FZE | Application Engineer Intern

Aug 2026 – Present

Jebel Ali Airport Freezone, Dubai, United Arab Emirates

- Building foundational expertise in Fisher control valves — valve types, actuators, positioners, and product lines — within a flow control and application engineering environment.
- Support specification, selection, and sizing workflows for control valve quotations, translating customer requirements into technical proposals.
- Develop Power Apps, Power Automate, and Power BI solutions to digitalize workflows across multiple departments, building sales platform applications and RFQ/project performance dashboards used company-wide.
- Analyze and visualize large-scale RFQ and Inside Sales datasets spanning thousands of records across multiple fiscal years, surfacing trends that inform business and sales decisions.
- Operate across mechanical systems, industrial automation, data analytics, and digital engineering, bridging technical and commercial teams.

Al Fisht Electromechanical Works L.L.C. | Electrical Engineer Intern

Apr 2026 – July 2026

Sheikh Zayed Road, Ajman, United Arab Emirates

- Diagnosed and resolved ELV/fibre-optic issues (splicing, cabling, damage rectification), performing OTDR testing and optical power measurements (EXFO, Fluke, AFL) across ISP, GSM, CCTV, and Intercom systems for construction and district cooling networks.
- Provided technical support and documentation across multiple active sites within a lean 4-person engineering team, preparing CRS, BOQ, Method Statements, TRA, Material Submittals, ISP wiring schedules, as-built drawings, and fibre routing layouts to ensure compliance with project specs and HSE requirements.
- Acted as primary client coordination point for major district cooling operators (Tabreed, Empower, Emicool, Emaar District Cooling), aligning on-site execution with in-office planning; completed Tabreed PTW/HSE authorization for plant environments.

Unique World Robotics L.L.C. | Robotics Intern

Sep 2025 - Jan 2026

Bur Dubai, Dubai, United Arab Emirates

- Delivered private, syllabus-based robotics and AI programming instruction to students aged 5–15 across the company's full curriculum — Arduino, LEGO Spike Prime/Essentials, Quarky, Raspberry Pi, Matatalab, DOBOT robotic arms, and TinkerCAD — adapting teaching style to each student's age and skill level.
- Coached a 5-student FIRST LEGO League (FLL) competition team (ages 14–16) through the full competition cycle — robot design, autonomous programming, and mission strategy — under FLL Challenge judging criteria.
- Served as the company's primary parent-facing spokesperson: marketed courses and programs to prospective families, communicated student progress, and resolved concerns with difficult parents — building trust and retention through clear technical and interpersonal communication.
- Built hands-on prototyping proficiency across Arduino, Raspberry Pi, ESP32, and micro:bit while translating technical robotics concepts into accessible lessons for young learners.

EDUCATION

B. Eng (Hons) Mechatronics Engineering | University of Wollongong

2022 - 2026

Dubai Knowledge Park, Dubai, United Arab Emirates

- Thesis: Pocket Wellness — AI-driven multimodal wearable glove for real-time physiological monitoring and mental health intervention.
- Relevant Coursework: Robotics, Control Systems, Mechatronics Design, Signal Processing, Embedded Systems, Machine Vision.

PROJECTS

PaveWay — Smart Road Defect Detection System

Sep 2024 - June 2025

Innovation Fair: 2nd Place

- Led a 6-person cross-functional team to design an autonomous ground robot with cameras, LiDAR, and AI-based image processing to detect and geospatially map road surface defects in real time.
- Built and validated the system on a purpose-built ~4 m² test rig simulating pavement defects, achieving 96%+ defect detection accuracy.
- Integrated sensor fusion to classify defect types and generate prioritized maintenance reports; built a data pipeline translating sensor data into GIS-tagged dashboards.
- Coordinated work across mechanical design, embedded programming, and AI modelling; awarded 2nd Place at the Innovation Fair.
- *Technologies: LiDAR, Camera Vision, Python, ROS, Gazebo, TurtleBot3, 3D Printing, SolidWorks, Embedded Systems, C++*

Pocket Wellness — AI Wearable Mental Health Platform

Sep 2025 - June 2026

Dissertation project graded with distinction

- Independently developed a wearable glove integrating GSR, HRV, EMG, and ECG biosensors on ESP32 to capture physiological signals linked to anxiety, depression, insomnia, and panic disorders.
- Designed a multi-modal AI classification engine (Random Forest with SHAP explainability) achieving 97.4% accuracy, triggering personalized interventions via chatbot and haptic feedback.
- Validated the model using physiological datasets sourced from Kaggle in place of live human trials, given ethical and access constraints on real-time participant testing; live-participant validation is scoped as future work.
- Built a full-stack analytics dashboard with privacy-first data handling; presented at the Innovation Fair and final thesis defense, graded with distinction.
- *Technologies: ESP32, GSR/HRV/EMG/ECG Biosensors, Python, Machine Learning: Random Forest, SHAP, Full-Stack Development*

SKILLS

CAD (SolidWorks, Fusion 360, AutoCAD, TinkerCAD)	Python	C/C++	Java	ROS2	ANSYS
OpenCV & Image Processing	Matlab (Simulink, Simscape, Simscape Multibody, Fuzzy Logic Toolbox)				
Machine Learning	Microsoft Suite (Power BI, Power Apps, Power Automate, Excel, SharePoint, PowerPoint)				
Microcontrollers & Embedded Systems (Arduino, ESP32, Raspberry Pi, micro:bit, AVR Assembly Programming)					
PLC Programming (Codesys)	OTDR Testing (EXFO, Fluke, AFL)		Full Function Valves (FF2) Softwares		

CERTIFICATIONS

- SAP S/4HANA Cloud Public Edition Consultant (SAP Official Certification)
- ROS2 – Jazzy (Udemy)
- PLC Programming (Udemy & PLCDojo)
- Understanding Artificial Intelligence (DataCamp), Introduction to AI Agents (DataCamp)

ADDITIONAL INFORMATION

Fluent in English and Malayalam, with conversational Hindi and basic French and Arabic. Comfortable working both independently and as part of a cross-functional team. Holds a valid UAE Driver's License.